# WAPT PHASE REPORT

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**Phase:** Phase 1

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## 1. Introduction

This report is the main summary of all tasks completed during WAPT Phase 1. The phase covered:

- Web Fundamentals
- Reconnaissance & Enumeration
- Burp Suite
- Common Web Vulnerabilities

Each task was completed with practical labs, evidence collection, and documentation.

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## 2. Task 1: Web Fundamentals

**Objective:** Understand how web applications work and HTTP communication.

**Rooms Completed:**

- TryHackMe: Web Application Basics
- TryHackMe: HTTP in Detail

**Key Concepts Learned:**

- Web application architecture (Client, Server, Database)
- HTTP/HTTPS protocols
- HTTP methods (`GET`, `POST`, `PUT`, `DELETE`)
- Headers, cookies, and sessions
- Status codes (`200`, `301`, `403`, `404`, `500`)
- Authentication basics
- Basic APIs

**Evidence:** Room completion screenshots, DevTools screenshots, Network tab, headers, cookies, status codes.

**Key Learnings:** HTTP is stateless; cookies maintain state; Developer Tools reveal all traffic; understanding HTTP is the foundation of web security.

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## 3. Task 2: Reconnaissance & Enumeration

**Objective:** Learn to map an application's attack surface.

**Rooms Completed:**

- TryHackMe: Passive Recon
- TryHackMe: Active Recon
- TryHackMe: Content Discovery

**Key Concepts Learned:**

- Passive vs Active Recon
- `WHOIS` and `DNS` lookups
- Subdomain discovery
- Directory brute forcing
- Technology identification
- Port scanning with `Nmap`
- Parameter and endpoint discovery

**Tools Used: `WHOIS`, `Nslookup`, `Nmap`, `Gobuster`, Browser DevTools.**

**Evidence:** Room completions, tool outputs (`WHOIS`, `DNS`, `Nmap`), directory scan results.

**Key Learnings:** Recon is the foundation of pentesting; passive leaves no trace; active is more thorough; enumeration reveals hidden attack surface.

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## 4. Task 3: `Burp Suite`

**Objective:** Learn to intercept, analyze, and modify web traffic.

**Room Completed:**

- TryHackMe: `Burp Suite Repeater`

**Key Concepts Learned:**

- `Burp Proxy` and interception
- `HTTP History`
- `Repeater` for manual testing
- `Intruder` for fuzzing
- `Decoder` for encoding/decoding
- Request/Response analysis

**Tools Used:** `Burp Suite Community Edition`, built-in browser.

**Evidence:** Proxy intercept, HTTP history, Repeater request/response, modified requests.

**Key Learnings:** `Burp Suite` is essential for web testing; `Repeater` allows precise manipulation; understanding traffic flow is critical.

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## 5. Task 4: Common Web Vulnerabilities

**Objective:** Learn to identify and test common web vulnerabilities.

**Labs Completed:**

- PortSwigger: `SQL Injection (WHERE clause)`
- PortSwigger: `Reflected XSS`
- PortSwigger: `Username Enumeration`
- TryHackMe: `IDOR`
- PortSwigger: `Path Traversal`

**Key Concepts Learned:**

- `SQL Injection`: unsanitized input in SQL queries
- `XSS`: unencoded output executing JavaScript
- Authentication flaws: username enumeration, weak passwords
- `IDOR`: accessing other users' data via ID manipulation
- Access Control: missing authorization checks
- `Path Traversal`: using `../` to read arbitrary files
- File Upload vulnerabilities

**Tools Used: `PortSwigger Academy`**, **`TryHackMe`**, **`Burp Suite (Intruder, Repeater)`**, **Browser DevTools**.

**Evidence:** Lab completions, payloads, responses, Burp Intruder results, unauthorized access proofs.

**Key Learnings:** Every vulnerability has a root cause and a fix; input validation and output encoding are critical; access control must be server-side.

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## 6. Overall Tools & Technologies Used

- TryHackMe
- PortSwigger Web Security Academy
- Burp Suite Community Edition
- Browser Developer Tools (F12)
- WHOIS, Nslookup, Nmap, Gobuster
- Google Chrome

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## 7. Overall Key Learnings

- Understanding HTTP is the foundation of web security
- Reconnaissance reveals the attack surface
- Burp Suite is essential for traffic analysis
- Every vulnerability has a root cause and remediation
- Practical hands-on labs reinforce theoretical concepts
- Documentation and evidence are as important as exploitation
- Security is about understanding "why", not just "how"

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## 8. Conclusion

Phase 1 provided a strong foundation in Web Application Penetration Testing. I completed all required TryHackMe rooms and PortSwigger labs, gathered evidence, and documented my learning. I am now prepared for Phase 2, where I will apply these skills to a real-world-style WAPT project.

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